

External Report on the PhD thesis of Arghya Datta

Thesis Decision: Accepted, with classification Excellent/Exceptional.

Summary: The thesis addresses the topic of probabilistic principal components analysis (PPCA) and aims to establish the statistical and computational properties of estimators and algorithms associated with this model. There are three main research chapters:

- Chapter 1 establishes the consistency of estimators of components of the PPCA model under independent sampling using a novel formulation.
- Chapter 2 introduces a robust Bayesian PPCA model and develops computational approaches to inference (fully Bayes and variational Bayes), as well as providing a commentary on identifiability issues.
- Chapter 3 rigorously establishes the desirable convergence properties of a variational inference algorithm for the PPCA model.

Specific comments on the research chapters:

- **Chapter 1:** *On the Consistency of Maximum Likelihood Estimation of Probabilistic Principal Component Analysis*

This Chapter introduces the PPCA model through Gaussian linear model formulation

$$X_i = \mathbf{W}Z_i + \epsilon_i \quad (1)$$

with $\epsilon \sim \text{Normal}(0, \sigma^2 \mathbf{I}_p)$. In this model $\mathbf{W}$ and σ^2 are the parameters of interest. The Chapter recognizes the inherent non-identifiability of this model, and develops the framework that allows the convergence of maximum likelihood (ML) estimates of these parameters to their data generating true values. The critical development is the realization that the notion of convergence, and the space and norm via which this convergence must be studied, is not the usual Euclidean space and norm, but instead a quotient space exploiting (essentially) the equivalence classes of parameter sets that yield the same marginal covariance

$$\Sigma = \mathbf{W}\mathbf{W}^\top + \sigma^2 \mathbf{I}_p.$$

Results are provided to establish the consistency of the ML estimator, which is available in closed form up to an arbitrary rotation matrix $\mathbf{R}$. These results are extended to the dependent data case, that is, where the observes x_i arise from some suitably behaved process that exhibits dependence. The Chapter concludes with an interesting discussion on the subtlety of the definition of the required topology to study the convergence, and some counterexamples to demonstrate the necessity of the elements of the formulation.

- **Chapter 2:** *A Bayesian robust approach to probabilistic PCA*

This chapter extends the model in (1) to allow for a heavy-tailed formulation. The scale mixture strategy is deployed where

$$X_i = \mathbf{W} + Z_i + U_i^{-1} \epsilon_i \quad (2)$$

with $U_i \sim \Gamma(\nu/2, \nu/2)$, thus yielding Student-t errors marginally. The chapter goes on to develop MCMC and coordinate ascent variational inference (CAVI) strategies to perform inference in the model. Theory is provided to establish the robust nature of the resulting model. Numerical experiments also demonstrate that MCMC methods can be effective but are computationally prohibitive when the sample size is large, and that CAVI has good and efficient performance as tractable approximation.

- **Chapter 3:** *On the convergence of CAVI for Bayesian PCA*

This Chapter studies the theoretical performance of the (widely used) CAVI algorithm specifically in the context of Gaussian-based PPCA. This convergence of the algorithm in the PPCA context is extremely challenging to study due to the high-dimensional and multi-modal nature of the objective function. The results here are impressive for such a complex setting: the conceptualization is quite straightforward, but the details are very extremely intricate, and the treatment comprehensive and convincing.

Each of these contributions is substantial in its own way. Chapter 1 is already published in NeuRIPS. I have no doubt that the other two chapters will be publishable in suitable venues.

For a classification of the thesis, I am on the fence between Excellent and Exceptional for the following reasons, that I hope to explore at the oral defense:

- (i) Chapter 1: It is unarguably the case that this model is non-identifiable for the individual components $\mathbf{W}_0$ and σ_0^2 – that is essentially the inspiration for the contribution – but that the quantity $\Sigma_0 = \mathbf{W}_0 \mathbf{W}_0^\top + \sigma_0^2 \mathbf{I}_p$ is identifiable as the (marginal) covariance of the observables. It is immediately evident, therefore, that a method of moments estimator would exhibit desirable convergence to the identifiable quantity. The theory provided in Chapter 1 establishes the convergence for the MLE using a quotient space formulation; however, the marginal likelihood is available, and I am not clear why a direct attack on the marginal likelihood and the quantity Σ_0 is insufficient. That is to say, what is the practical utility of establishing convergence in the correct but more abstract space/metric? Also, suppose that more than just consistency but also some form of uncertainty representation is required; what does the formulation deployed offer?
- (ii) Chapter 2: the extension to a multivariate-t model exploiting the scale mixture approach, and the consequent computational strategies developed, are useful but reasonably straightforward. However, having read the intricate formulation and detailed treatment in Chapter 1, using the quotient space formulation, I was anticipating some of this to be deployed in Chapter 2, perhaps to improve the MCMC implementation (eg moves exploiting the quotient space formulation). I think there is potentially a link that could have been investigated and exploited; as some of the numerical results show, Gibbs moves in the latent space formulation are laborious and ripe for improvement. I will confess that I am not wholly convinced in general by the concept of variational approximation in models that are amenable to MCMC, but the context here demonstrates why they can be advantageous.
- (iii) The presentation of the thesis could be improved as there are a number of anomalous elements, and I will provide a marked-up copy with some suggested amendments. Despite this, I would note that the mathematical presentation of the concepts is very good throughout the thesis.

Overall, the contribution of the thesis is of high quality, and the student is to be congratulated.

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External Examiner
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